

2026 Pleasanton Math Tournament

Countdown Round

What is the median of the positive perfect cubes less than 1000?

125

A super-prime is a prime number whose position in the sequence of primes is itself a prime number. For example, 67 is a super-prime because it is the 19th prime, and 19 is prime. What is the third super-prime?

Shreyas has 100 dollars to buy ice-cream sandwiches and popsicles. Each ice-cream sandwich costs 6 dollars and each popsicle costs 4 dollars. What is the maximum amount of popsicles that he can buy if he already bought 4 ice-cream sandwiches?

What is the largest base 10 number that can be expressed as a 4-digit base 5 number?

624

What is the last digit of 19^{2026} ?

What is the largest prime factor of 351?

Vatsal is listing all the numbers less than 20 which are multiples of 2 but not multiples of 3. How many numbers does he list?

Given that the volume of sphere A is 27 times the volume of sphere B, how many times larger is the surface area of sphere A compared to sphere B?

Arnav starts counting all the positive integers between 1 and 100. However, he skips any composite number. How many numbers are between 1 and 100 according to Arnav?

If $p + m = 29$, $m + t = 46$, and $p + t = 53$, what is the value of $p + m + t$?

What is the area, in square units, of a square whose side length is equal to the radius of a circle with circumference 20π units?

What are the last two digits of 5^{6^7} ?

25

How many digits are in the base-ten representation of $32^7 * 5^{29} * 175^3$?

Where does the point $(2, 4)$ land after a rotation clockwise 90 degrees about the origin followed by a reflection across the y-axis?

$(-4, -2)$

Triangle ABC has side lengths $AB = 25$, $BC = 60$, and $AC = 65$. What is the average, in degrees, of its three internal angles?

The area of three concentric circles is drawn such that the area of the smallest circle is equal to the area between the two larger circles. If the area of the largest circle is 100π and the area of the smallest circle is 36π , what is the radius of the middle circle?

What is the least positive perfect square that is a multiple of 45?

225

What is the smallest positive integer that has exactly 11 unique positive divisors?

1024

How many positive factors of $26!$ are prime numbers?

A 9-sided die has faces numbered 1 through 9. One of the 45 total dots is randomly colored blue and the die is rolled twice. What is the probability that the blue dot appeared on the top face both times?

$$1/81$$

Vatsal writes down three consecutive positive integers. He erases the middle number and multiplies the remaining two numbers to get 255. What was the middle number that he erased?

A convex hexagon has three right angles. The remaining three interior angles are all congruent. What is the measure, in degrees, of one of these congruent angles?

Triangle ABC has side lengths 9, 40, and 41, with hypotenuse AC. A circle with center O passes through all three vertices of the triangle. What is the length of BO?

20.5 (accept $41/2$)

How many positive integer divisors does 2026^{3^2} have?

A bag contains 3 red marbles and 7 blue marbles. Marbles are drawn one at a time without replacement until a red marble is drawn. What is the probability that the second marble drawn is the first red marble?

7/30

If N is the number of positive divisors of $\frac{26!}{20!}$, what is the sum of the digits of N ?

In rhombus $ABCD$, each side has length 10 cm. Diagonal BD has length 10 cm. Find the area of rhombus $ABCD$. Express your answer in the simplest radical form.

$$50\sqrt{3}$$

What is the sum of the largest and second-largest prime factors of 2026?

1015

On the whiteboard, Shreyas writes down every divisor of 2025. What is the sum of the numbers he wrote?

3751

Four people are sitting in a row at a movie theater. They all go to the restroom and sit back down such that none of them sits in their previous seat. How many ways can they reseal themselves?

After the year 2026, when will the next palindrome year be?

2112

What is the least positive integer divisible by 4 distinct prime numbers?

210

For what positive value of x is $(x - 5)(x - 6) = 30$?

Shreyas flips 2025 coins. What is the probability that he gets more heads than tails?

$$1/2$$

If January 1st, 2026 fell on a Thursday, what day of the week will June 1st, 2026 fall on?

Monday

In how many ways can 5 identical trophies be awarded to 5 students?

Three consecutive positive integers a , $a + 1$, and $a + 2$ are written on a paper. Find the positive difference between the product of the smallest and largest integers and the square of the middle integer.

What is the value of $2! + 0! + 2! + 6!$?

725

How many ways are there to arrange the letters PMTPMT?

What is the value of $47 * 53$?

2491

How many diagonals does a dodecagon (12 sided polygon) have?

A box contains more than 100 coins. It can be divided among 4, 6, and 9 bags with one coin left over. What is the smallest number of coins that could possibly be in the box?

Vatsal walks up a hill at 5 mph, and bikes down at 20 mph. What was his average speed throughout the journey?

8 mph

In a magical land far far away, there are 4 feet in a liter, 10 liters in a grams, and 3 grams in a kelvin. How many feet are in 2 kelvin?

240

Next year when PMT expands, the top 255 out of 2000 students are selected to play the countdown round, with the top seed being given a bye for the first round. How many matches will be played in total to crown a Countdown Round Champion?

A number is chosen at random from the integers 1 through 30. Given that the number is divisible by 3, what is the probability that it is also divisible by 5?

$$1/5$$

Arnav writes the number 2^{2026} on a paper. How many digits does he write, given that $10^{0.301} = 2$?

Athithan wrote down some number of consecutive integers. The sum of the integers is 190. What is the maximum number of integers Athithan could've written?

In the game “Bubble Sort,” Arnav must reorder the array $(10, 9, 8, 7, 6, 5, 4, 3, 2, 1)$ into $(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$ to win. However, he can only perform “bubble switches,” which switch two consecutive elements. For example, changing $(4, 7)$ into $(7, 4)$. What is the least number of bubble switches Arnav must perform to win the game?

At Math High School, students may enroll in Calculus, Number Theory, and Statistics. Each student takes at least two different courses. Out of 100 students total, 40 are enrolled in both Calculus and Number Theory, 30 in both Calculus and Statistics, and 50 in both Number Theory and Statistics. How many students are enrolled in all three courses?

Evaluate the sum $1 + 2 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 + 2^8 + 2^9$?

1023

If the number PMT is equal to 93_{16} where P, M , and T are digits in base 10, what is $P + M + T$?

Tiger has 3 cubes of side length 1, 2, and 3 each. To create a sturdy tower, only smaller cubes can go on top of bigger cubes. How many possible towers can be made using at least 1 cube?

Isha can complete an essay in 6 hours. Isha and Taiba together can complete an essay in 3 hours. How many hours does Taiba need to complete an essay?

How many non-congruent equilateral triangles with integer side lengths have area less than or equal to $64\sqrt{3}$?

An arithmetic sequence has 13 terms. If the first and last terms are 53 and 14 respectively, what is the common difference of the sequence?

$$-13/4$$

What is the area of an isosceles right triangle with perimeter $4 + 4\sqrt{2}$ inches?

The largest isotope of unobtainium is radioactive and loses half its mass every year. If we have 1 kg of the sample left after keeping it for exactly 5 years, how much mass did the sample originally have?

What is the unit digit of the product of the first 9 positive square numbers?

0

Tiger divides his favorite number x by 17 and gets a remainder of 7. What is the remainder Tiger gets when he divides x^2 by 17?

Athithan has a goal of benching 400 lbs. He can currently bench 200 lbs. His personal record increases by 10 lbs each week for the first 5 weeks, after which it starts increasing by 15 lbs. In how many weeks will Athithan achieve his goal of benching 400 lbs?

Isha is picking an outfit to wear! She has 9 colors each for pants, shirts, and sweatshirts. She does not want to wear an outfit that has exactly two items of clothing with matching colors. How many outfits can she make?

In an English test, there are 5 true or false questions. Tiger needs to get every question correct in order to pass the class. What is the chance that this happens if Tiger gets the 1st question correct, but guesses the rest?

The double factorial is defined as the following: If n is even, $n!!$ is the product of all the even positive integers less than or equal to n . If n is odd, $n!!$ is the product of all the odd positive integers less than or equal to n . What is the value of $10!!$?

3840

A polygon has 26 sides. What is the sum of all of its exterior angles?

360

Medha and Arnav are having a 200-yard swimming race. Medha swims 25 yards in 12 seconds, and gets 1.5 seconds slower every 25 yards. Arnav swims 50 yards in 25 seconds, and gets 5 seconds slower every 50 yards. How many seconds does the winner win by?

What is the maximum number of points of intersection between a hexagon and a circle?

12

Two numbers add up to -12 and their difference is 20. If n is the larger of the two numbers, what is $n!$?

A perfect number is a number equal to the sum of all its proper divisors. What is the 2nd perfect number?

When $100!$ is written in base 10, how many zeros does it have at the end?

Athithan sums up every number of Pascal's triangle from row 0 to row 10. If the sum of his numbers can be written in the form $a^b + c$ where $a > 1$, $c > 1$ and the value of b is maximized. What is the value of $a + b + c$?

Find the last two digits of the sum $21^{2026} + 31^{2026}$

Isha computed the sum of the first n positive integers, but accidentally counted one integer twice. Her final total was 189. What integer was counted twice?

What is the remainder when the sum $2026^3 + 2026^2 + 2026^1 + 2026^0$ is divided by 6?

Two fair six-sided dice each have faces numbered 1, 2, 2, 4, 7, and 9. When both dice are rolled, how many distinct sums are possible?

The digits of 6147 are rearranged to create a new four-digit integer. What is the smallest possible positive difference between this new integer and the original number 6147?